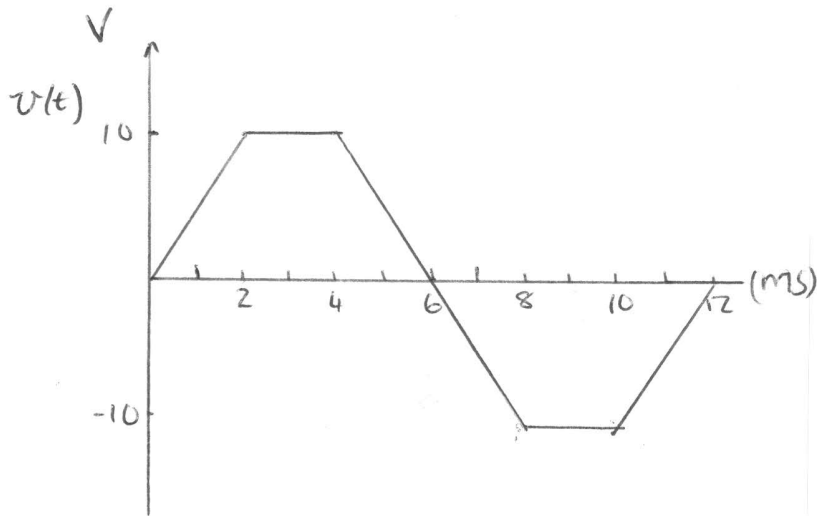


Tutorial 7

6.6 *



* $C = 55 \mu\text{F}$

* Draw the current waveform.

* $i = C \left(\frac{dv}{dt} \right)$ → slope of waveform.

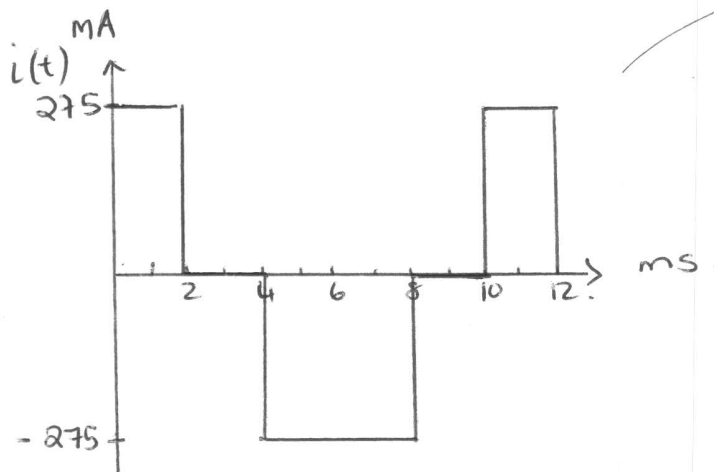
* ex; for $0 < t < 2$:

$$\frac{dv}{dt} = \frac{10 - 0}{2 \times 10^{-3} - 0} = 5000$$

$$\Rightarrow i = C \frac{dv}{dt} = 55 \mu (5000) = \underline{275 \text{ mA}}$$

* Max/min current.

∴ Current waveform :



* From $i = C \frac{dv}{dt}$

→ current flows through capacitor while

$$\frac{dv}{dt} \neq 0$$

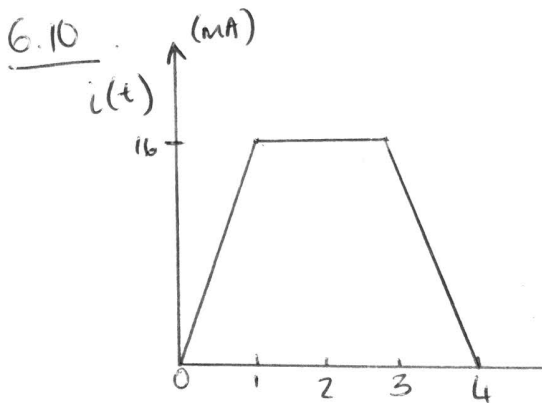
$$\therefore \text{if } \frac{dv}{dt} = 0 \Rightarrow i(t) = 0$$

6.7. * @ $t=0 \rightarrow V(0) = 10 \text{ V}$

* $C = 25 \text{ mF}$

* $i(t) = 5t \text{ mA}$

$$\begin{aligned} V &= \frac{1}{C} \int i \, dt + V(t_0) \\ &= \frac{1}{25 \times 10^{-3}} \left[\int_0^t (5t) \times 10^{-3} \, dt + 10 \right] \\ &= \frac{2.5t^2}{25} + 10 \\ &= (0.1t^2 + 10) \text{ V} \end{aligned}$$



* $C = 5 \text{ mF}$

* Determine current through capacitor.

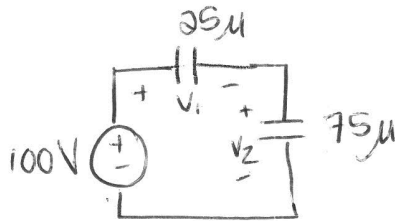
$$\Rightarrow i(t) = C \frac{dv}{dt} = 5 \times 10^{-3} \frac{dv}{dt}$$

$$* V = \begin{cases} 16t, & 0 \leq t < 1 \mu\text{s} \\ 16, & 1 \leq t < 3 \mu\text{s} \\ 64 - 16t, & 3 \leq t < 4 \mu\text{s} \end{cases}$$

$$* \frac{dv}{dt} = \begin{cases} 16 \mu, & 0 \leq t < 1 \\ 0, & 1 \leq t < 3 \\ -16 \mu, & 3 \leq t < 4 \end{cases}$$

$$\Rightarrow i(t) = \begin{cases} 80 \text{ nA}, & 0 \leq t < 1 \mu\text{s} \\ 0, & 1 \leq t < 3 \mu\text{s} \\ -80 \text{ nA}, & 3 \leq t < 4 \mu\text{s} \end{cases}$$

6.15



1) * series *

$$\begin{aligned} \bullet C_{eq} &= \frac{1}{C_1} + \frac{1}{C_2} \\ &= \left(\frac{1}{25\mu} + \frac{1}{75\mu} \right) \\ &= 18,75\mu F \end{aligned}$$

$$\bullet W = \frac{1}{2} C V^2$$

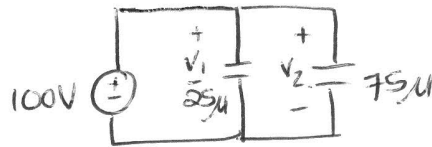
$$\Rightarrow V_1 = 100 \left(\frac{75}{100} \right) = 75V$$

$$V_2 = 25V$$

$$\begin{aligned} \therefore W_{25} &= \frac{1}{2} (25\mu) (75^2) \\ &= 70,3 \text{ mJ} \end{aligned}$$

$$\begin{aligned} W_{75} &= \frac{1}{2} (75\mu) (25^2) \\ &= 23,44 \text{ mJ} \end{aligned}$$

* Find energy stored *



2) * parallel *

$$\begin{aligned} \bullet C_{eq} &= C_1 + C_2 \\ &= 25\mu + 75\mu \\ &= 100\mu F \end{aligned}$$

$$\bullet W = \frac{1}{2} C V^2$$

$$\begin{aligned} \Rightarrow V_1 &= 100V \\ V_2 &= 100V \end{aligned}$$

$$\begin{aligned} \therefore W_{25} &= \frac{1}{2} (25\mu) (100^2) \\ &= 125 \text{ mJ} \end{aligned}$$

$$\begin{aligned} W_{75} &= \frac{1}{2} (75\mu) (100^2) \\ &= 375 \text{ mJ} \end{aligned}$$

* Note *

- For cap's in series, voltage across cap can be calculated as follows:

$$\Rightarrow q = C_{eq} V$$

$$\therefore q = (18,75\mu) (100) = 0,001875$$

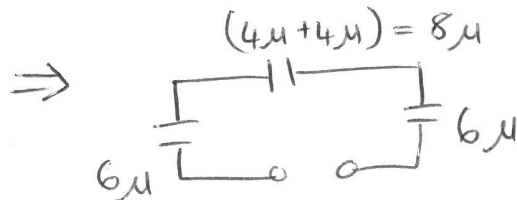
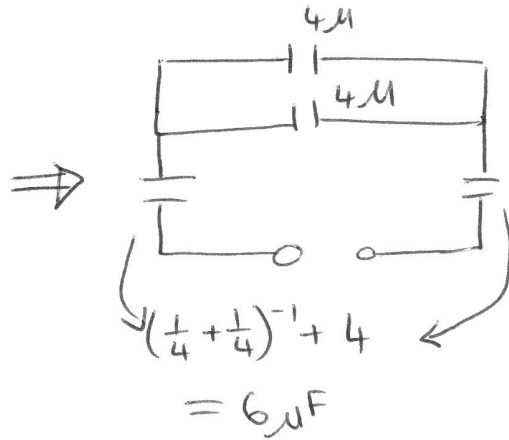
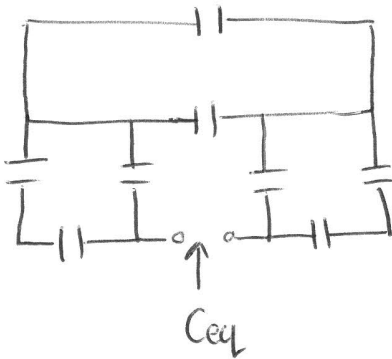
$$\Rightarrow V_1 = \frac{q}{C_1} = 75V$$

$$V_2 = \frac{q}{C_2} = 25V$$

6.18.

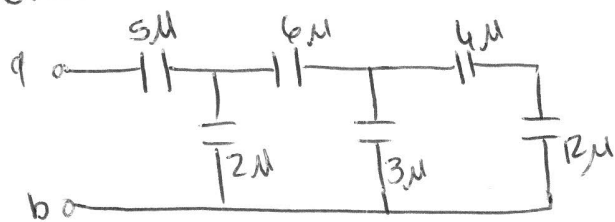


Find C_{eq} if all capacitors are $4\mu F$.



$$\therefore C_{eq} = \left(\frac{1}{6} + \frac{1}{8} + \frac{1}{6} \right)^{-1} = 2.18\mu F$$

6.21



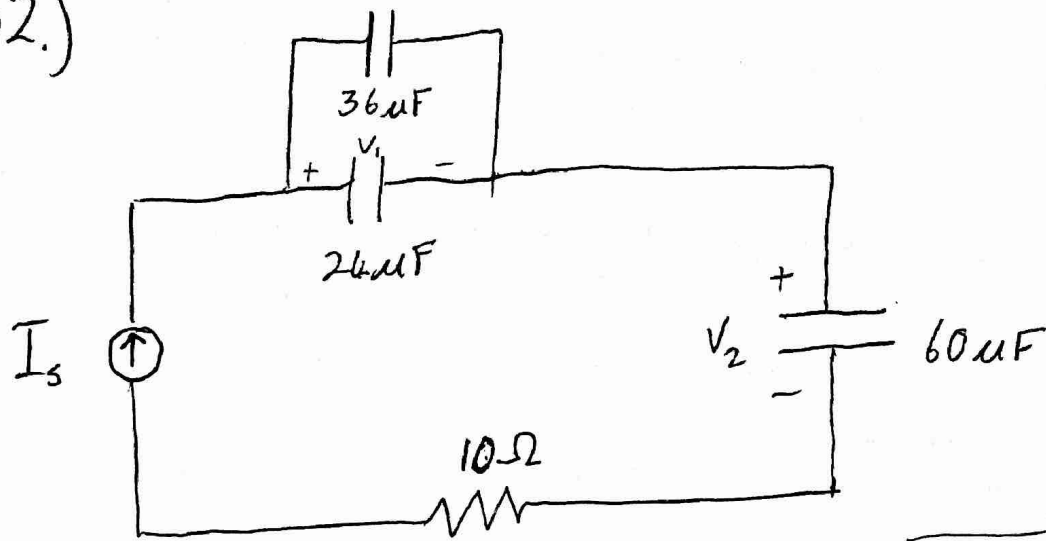
Determine C_{eq} at a-b.

$$C_{eq1} = \left(\frac{1}{4} + \frac{1}{12} \right)^{-1} + 3\mu = 6\mu$$

$$C_{eq2} = \left(\frac{1}{6} + \frac{1}{6} \right)^{-1} + 2\mu = 5\mu$$

$$C_{eq} = \left(\frac{1}{5\mu} + \frac{1}{5\mu} \right)^{-1} = 2.5\mu F$$

6.32.)



$$I_s = 4,5 e^{-2t} \text{ mA}$$

$$V_1(0) = 0 \text{ V}$$

$$V_2(0) = 0 \text{ V}$$

a.)
$$v(t) = \frac{1}{C} \int_{t_0}^t i dt + v(t_0)$$

$$V_1(t) = \frac{1}{(36\mu + 24\mu)} \int_{t_0}^t (4,5 e^{-2t}) dt + 0$$

$$V_1(t) = \frac{1}{60\mu} \left[-\frac{4,5}{2} e^{-2t} \right]_0^t$$

$$V_1(t) = \frac{1}{60\mu} \left[-\frac{9}{4} e^{-2t} + \frac{9}{4} \right]$$

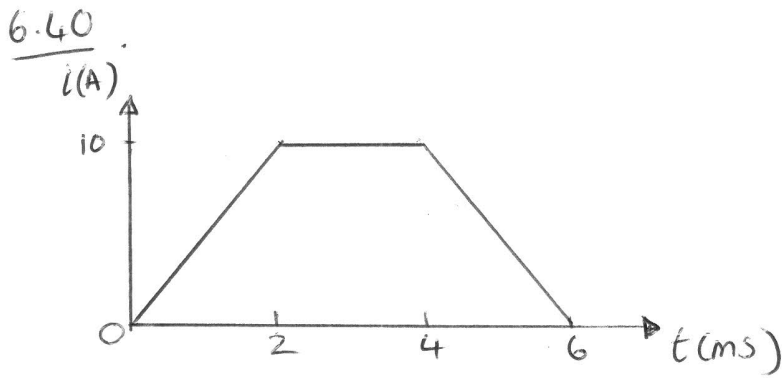
$$V_1(t) = 37500(1 - e^{-2t}) \text{ mV} \Rightarrow \underline{\underline{37,5(1 - e^{-2t}) \text{ V}}}$$

$$\underline{\underline{V_1(t) = V_2(t)}}$$

b.) Energy $\Rightarrow$
$$W = \frac{1}{2} C V^2$$

$$W_{36} = \frac{1}{2} (36 \times 10^{-6}) (37,5)^2 = 25,31 \text{ mJ}$$

$$W_{60} = \frac{1}{2} (60 \times 10^{-6}) (37,5)^2 = 42,19 \text{ mJ} \quad W_{24} = \frac{1}{2} (24 \times 10^{-6}) (37,5)^2 = 16,88 \text{ mJ}$$



* $L = 5 \text{ mH}$

* Determine voltage at $t = 1, 3, 5 \text{ ms}$.

*
$$U = L \frac{di}{dt}$$

$$i = \begin{cases} 5t & , 0 < t < 2 \\ 10 & , 2 < t < 4 \\ 30 - 5t & , 4 < t < 6 \end{cases}$$

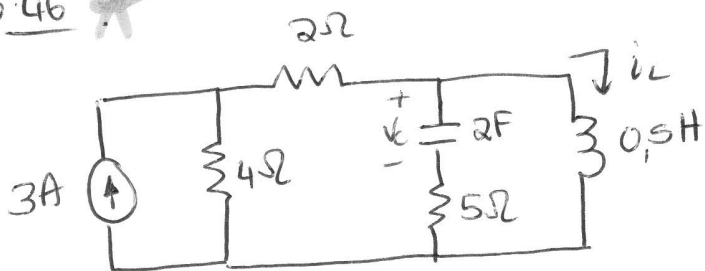
$$\frac{di}{dt} = \begin{cases} 5000 & , 0 < t < 2 \\ 0 & , 2 < t < 4 \\ -5000 & , 4 < t < 6 \end{cases}$$

$$\therefore @ t = 1 \text{ ms}: V = 5 \text{ m}(5000) = \underline{25 \text{ V}}$$

$$@ t = 3 \text{ ms}: V = 5 \text{ m}(0) = \underline{0 \text{ V}}$$

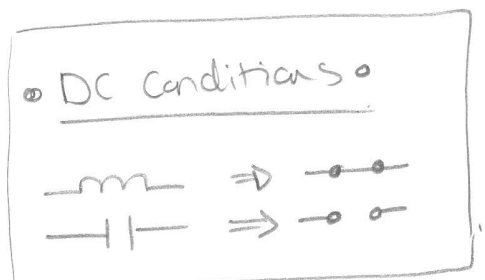
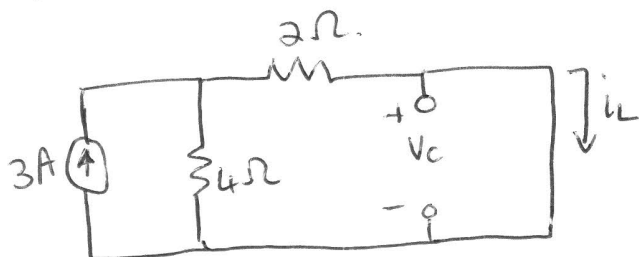
$$@ t = 5 \text{ ms}: V = 5 \text{ m}(-5000) = \underline{-25 \text{ V}}$$

6.46 *



* Find V_c , i_L and energy stored in capacitor and inductor under dc conditions

* Under dc-conditions:



top.

$$\rightarrow \bullet i_{2\Omega} = \left(\frac{4}{4+2} \right) 3 = 2A \quad (\text{Current division})^*$$

$$\therefore i_L = i_{2\Omega} = \underline{2A}$$

$$\bullet V_C = 0V \quad (\text{all current through SC})^*$$

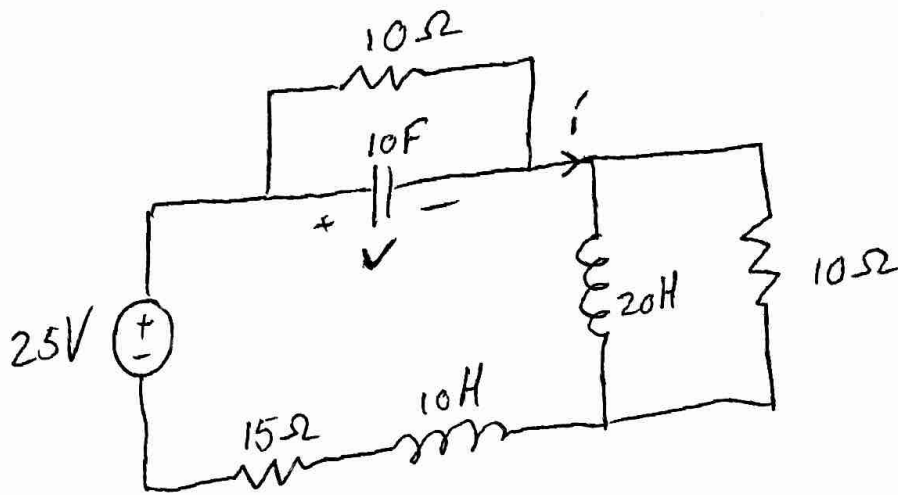
$$\rightarrow \boxed{W_L = \frac{1}{2} L i^2}$$

$$\therefore W_L = \frac{1}{2} (0.5) (2^2) \\ = \underline{1J}$$

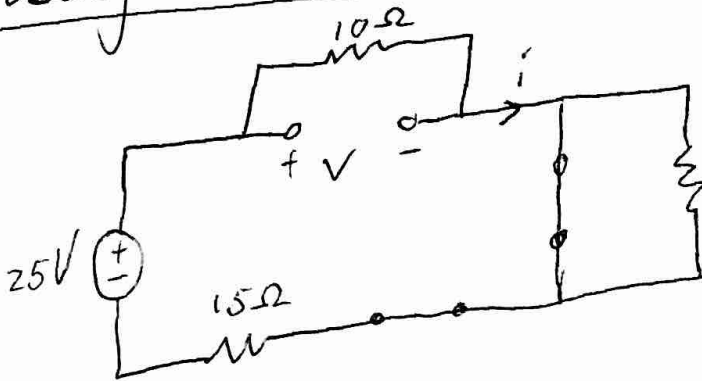
$$\boxed{W_C = \frac{1}{2} C v^2}$$

$$W_C = \frac{1}{2} (2) (0^2) \\ = \underline{0J}$$

6,48.)



Steady state conditions:



$$\underline{\underline{i:}} \quad R = \frac{V}{I}$$

$$(0+15) = \frac{25}{i}$$

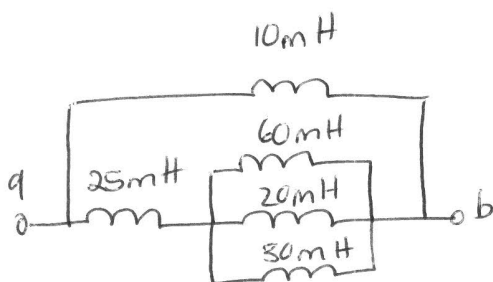
$$\underline{\underline{i = 1A}}$$

$$\underline{\underline{V:}} \quad \underline{\underline{\text{Voltage division:}}}$$

$$V = 25 \left(\frac{10}{10+15} \right)$$

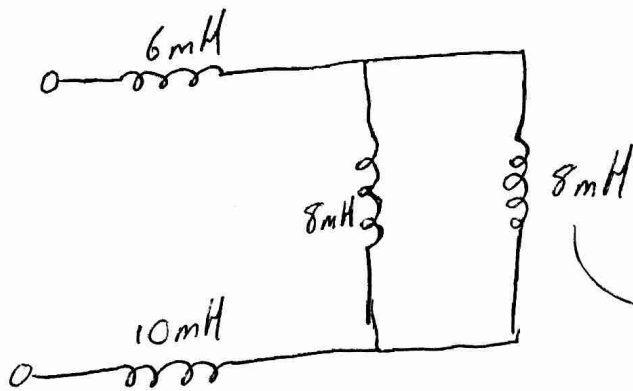
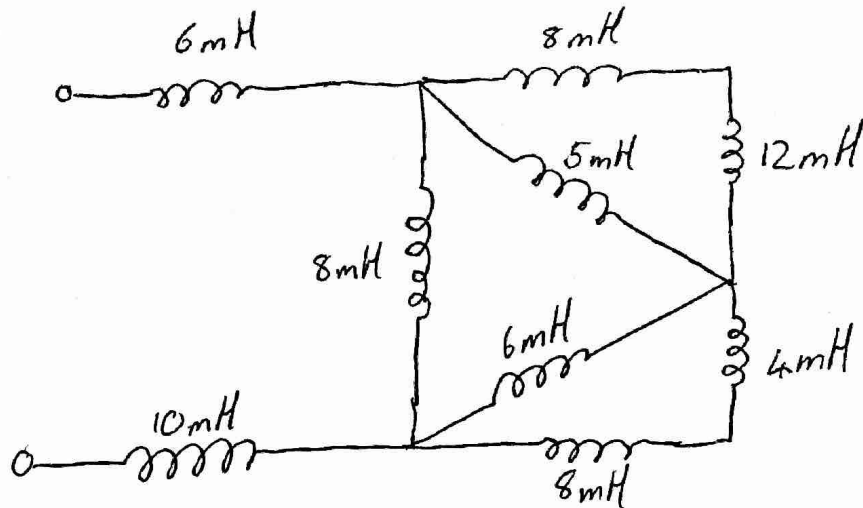
$$\underline{\underline{V = 10V}}$$

6.51.



$$\begin{aligned}\Rightarrow L_{eq} &= 10\text{m} \parallel (25\text{m} + 60\text{m} \parallel 20\text{m} \parallel 30\text{m}) \\ &= 10\text{m} \parallel (25\text{m} + 10\text{m}) \\ &= \underline{7,778\text{mH}}\end{aligned}$$

6.53.)



① $(8+12) \parallel 5 = 4mH$

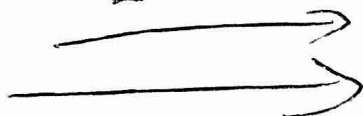
② $(8+4) \parallel 6 = 4mH$

③ $4+4 = 8mH$

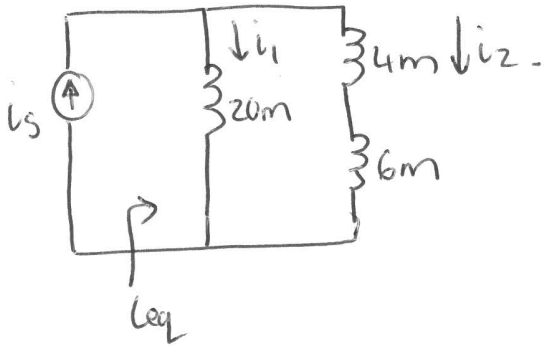
$$L_{eq} = 6 + 8 \parallel 8 + 10$$

$$= 6 + 4 + 10$$

$$= 20mH$$



6.61



a) L_{eq} , $i_1(t)$, $i_2(t)$ if
 $i_s = (3e^{-t})\text{mA}$

b) $v_o(t)$

c) Energy stored in 20mH inductor
 at $t=1\text{s}$

a) • $L_{eq} = 20\text{m} \parallel (4\text{m} + 6\text{m})$
 $= 6,667\text{ mH}$ →

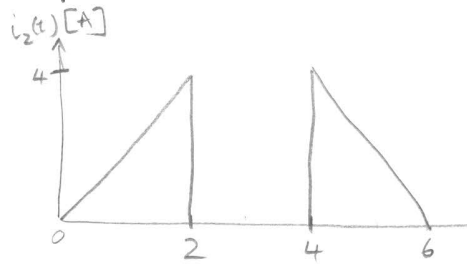
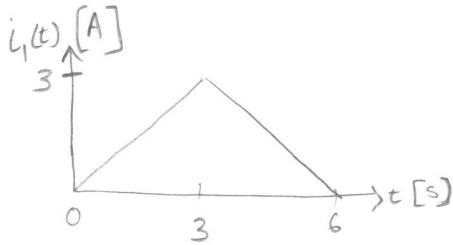
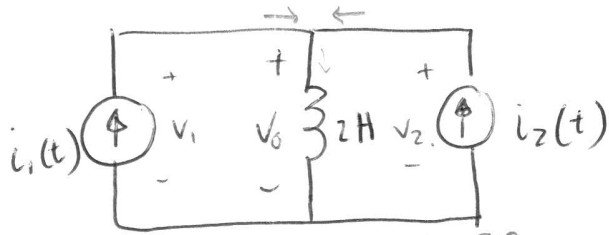
• $i_1(t) = i_s \left(\frac{10\text{m}}{20\text{m} + 10\text{m}} \right) = \frac{3}{3} e^{-t} = e^{-t}\text{mA}$ →

• $i_2(t) = i_s - i_1(t) = 2e^{-t}\text{mA}$ →

b) $v_o = L_{eq} \frac{di}{dt} = 6,667\text{m} (-3e^{-t})\text{m.}$
 $= -20e^{-t}\text{ }\mu\text{V}$ →

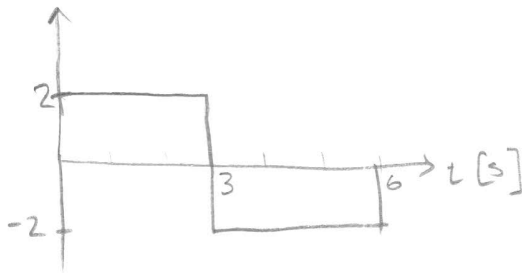
c) $w = \frac{1}{2} L i_1^2$
 $= \frac{1}{2} (20\text{m}) (e^{-t})^2$ @ $t=1\text{ms.}$
 $= 1.3534\text{ nJ}$ →

6.63



$$\rightarrow V_0 = V_1 + V_2$$

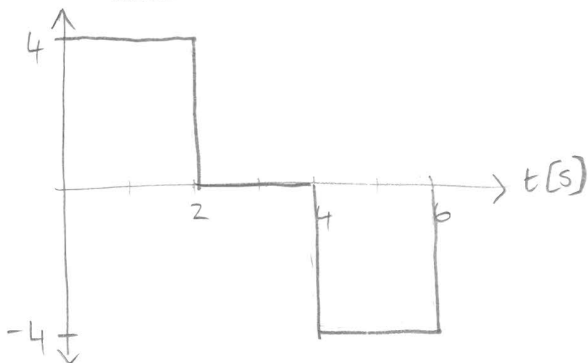
* V_1



$$* V = L \frac{di}{dt}$$

$$= 2 \times \begin{cases} 1 & , 0 < t < 3 \\ -1 & , 3 < t < 6 \end{cases}$$

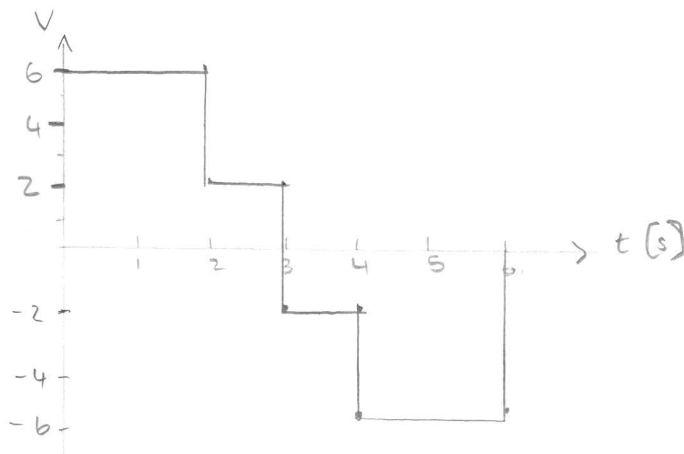
* V_2



$$* V = L \frac{di}{dt}$$

$$= 2 \begin{cases} 2 & , 0 < t < 2 \\ 0 & , 2 < t < 4 \\ -2 & , 4 < t < 6 \end{cases}$$

$$\Rightarrow \underline{V_0 = V_1 + V_2}$$



9.3.) Express the functions in cosine form

$$\begin{aligned} \text{a.) } 10 \sin(\omega t + 30) \\ &= (10) \cos(\omega t + 30 - 90) \\ &= \underline{10 \cos(\omega t - 60)} \end{aligned}$$

$$\begin{aligned} \text{b.) } -9 \sin(8t) \\ &= \underline{-9 \cos(8t + 90)} \end{aligned}$$

$$\begin{aligned} \text{c.) } -20 \sin(\omega t + 45) \\ &= -20 \cos(\omega t + 45 + 90) \\ &= \underline{-20 \cos(\omega t + 135)} \end{aligned}$$

$$\begin{aligned} \text{9.5.)}^* V_1 &= 45 \sin(\omega t + 30^\circ) \text{ V} \\ V_2 &= 50 \cos(\omega t - 30^\circ) \text{ V} \end{aligned}$$

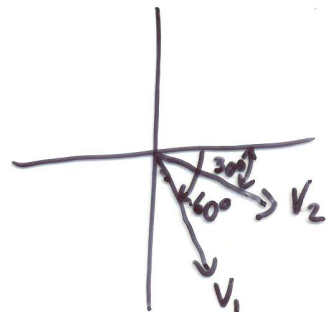
Determine the phase angle, also which one lags the other?

① Convert sine to cosine:

$$\begin{aligned} V_1 &= 45 \sin(\omega t + 30^\circ) = 45 \cos(\omega t + 30 - 90) \\ &= \underline{45 \cos(\omega t - 60)} \end{aligned}$$

$$\underline{V_2 = 50 \cos(\omega t - 30^\circ)}$$

$\therefore V_1$ lags V_2 by 30°



9.6) For the following pairs, determine which one leads and by how much.

a.) $v(t) = 10 \cos(4t - 60^\circ)$

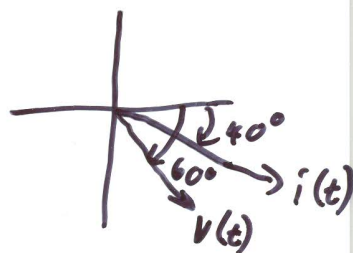
$$i(t) = 4 \sin(4t + 50^\circ)$$

① Convert to cosine form:

$$v(t) = 10 \cos(4t - 60^\circ)$$

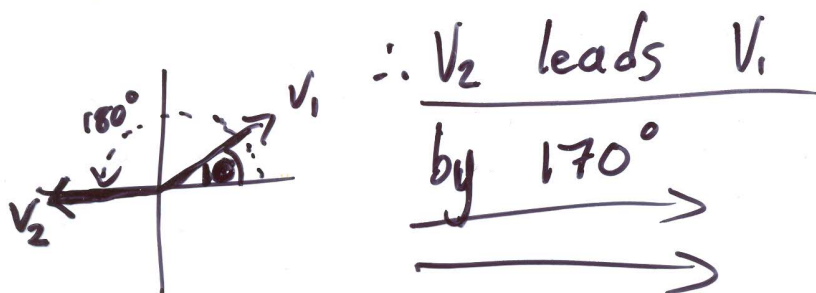
$$i(t) = 4 \sin(4t + 50^\circ) = 4 \cos(4t + 50^\circ - 90^\circ) \\ = 4 \cos(4t - 40^\circ)$$

$\therefore i(t)$ leads $v(t)$ by 20°



b.) $V_1 = 4 \cos(377t + 10^\circ)$

$$V_2 = -20 \cos(377t) = 20 \cos(377t + 180^\circ)$$



$$c.) \quad x(t) = 13 \cos 2t + 5 \sin 2t$$

$$y(t) = 15 \cos(2t - 11,8)$$

① Convert to cosine form:

$$x(t) = 13 \cos 2t + 5 \sin 2t$$

$$= 13 \cos 2t + 5 \cos(2t - 90)$$

$$= 13 \angle 0^\circ + 5 \angle -90^\circ$$

* [Use polar form to add the two terms.]

$$= 13 + (5 \cos(-90) + j 5 \sin(-90))$$

$$= 13 + 0 - j5$$

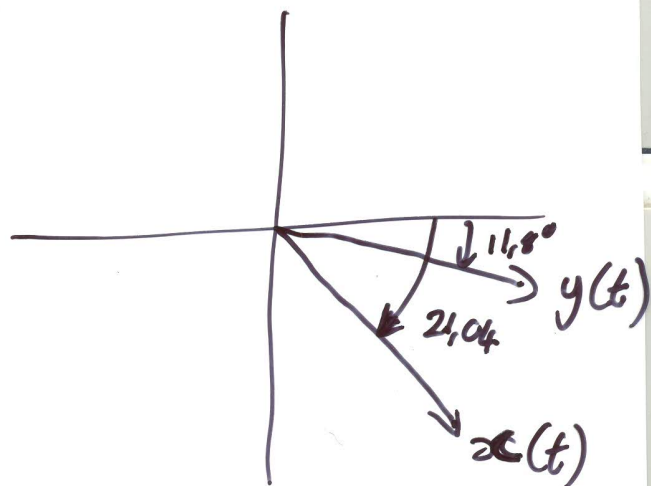
$$= \sqrt{(13)^2 + (-5)^2} \angle \tan^{-1}\left(\frac{-5}{13}\right)$$

$$= 13,93 \angle -21,04^\circ = 13,93 \cos(2t - 21,04^\circ) \rightarrow$$

$$\underline{y(t) = 15 \cos(2t - 11,8)} \rightarrow$$

$\therefore y(t)$ leads $x(t)$

by $9,24^\circ \rightarrow$



9.9.) * Evaluate the following complex numbers:
(Leave in polar form)

$$a.) 5 \angle 30^\circ (6 - j8 + \frac{3 \angle 60^\circ}{2 + j})$$

$$= 5 \angle 30^\circ (6 - j8 + \frac{3 \angle 60^\circ}{\sqrt{5} \angle 26,57^\circ})$$

$$= 5 \angle 30^\circ (6 - j8 + (\frac{3}{\sqrt{5}}) \angle 33,43^\circ)$$

$$= 5 \angle 30^\circ (6 - j8 + 1,12 + j0,74)$$

$$= 5 \angle 30^\circ (-7,12 - j7,26)$$

$$= 5 \angle 30 (10,17 \angle -45,56)$$

$$= 50,84 \angle -15,51^\circ$$



$$b.) \frac{(10 \angle 60^\circ)(35 \angle -50^\circ)}{(2+j6) - (5+j)}$$

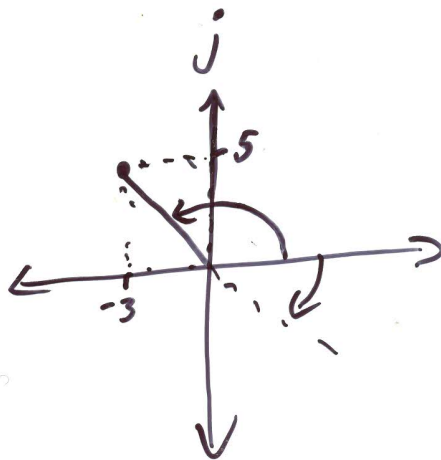
$$= \frac{350 \angle 10^\circ}{-3 + j5}$$

$$= \frac{350 \angle 10^\circ}{5,83 \angle 120,96^\circ}$$

$$= 60,02 \angle -110,96^\circ$$

→

→



9.19.) * Use phasors to find:

a.) $3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$

$$= 3 \angle 10^\circ - 5 \angle -30^\circ$$

$$= 2,95 + j0,52 - 4,33 + j2,5$$

$$= -1,38 + j3,02 = 3,32 \angle 114,49^\circ$$

~~$= 3,32 \angle 68,44^\circ$~~

b.) $40 \sin(50t) + 30 \cos(50t - 45^\circ)$

$$= 40 \cos(50t - 90^\circ) + 30 \cos(50t - 45^\circ)$$

$$= 40 \angle -90^\circ + 30 \angle -45^\circ$$

$$= 0 - 40j + 21,21 - j21,21$$

$$= 21,21 - j61,21 = 64,79 \angle -70,89^\circ$$

9.23.) Apply phasor analysis to evaluate the following.

a.) $v = [110 \sin(20t + 30) + 220 \cos(20t - 90)] V$

$$v = 110 \sin(20t + 30 - 90) + 220 \cos(20t - 90)$$

$$v = 110 \angle -60^\circ + 220 \angle -90$$

$$v = 55 - j95,26 + (220)(0) - 220j$$

$$v = 55 - j315,26$$

$$v = 320,02 \angle -80,10^\circ$$

$$v = 320,02 \cos(20t - 80,10^\circ)$$



$$b.) i = [30 \cos(5t + 60) - 20 \sin(5t + 60)] A$$

$$i = 30 \cos(5t + 60) - 20 \cos(5t + 60 - 90)$$

$$i = 30 \angle 60 - 20 \angle -30$$

$$i = 15 + j25,98 - 17,32 + j10$$

$$i = -2,32 + j35,98$$

$$i = 36,05 \angle (180 + (-86,31))$$

$$i = 36,05 \angle 93,68^\circ$$

$$i = 36,05 \cos(5t + 93,68^\circ)$$

$$\begin{aligned}
c.) & 20 \sin 400t + 10 \cos(400t - 60^\circ) - 5 \sin(400t - 20^\circ) \\
&= 20 \cos(400t - 90^\circ) + 10 \cos(400t - 60^\circ) - 5 \cos(400t - 110^\circ) \\
&= 20 \angle -90^\circ + 10 \angle 60^\circ ~~and 110^\circ~~ - 5 \angle -110^\circ \\
&= -j20 + 5 + j8,66 + 1,71 + j4,69 \\
&= 6,71 - j6,64 \\
&= 9,44 \angle -44,7^\circ
\end{aligned}$$